MindGuard

Week 2 Deliverables Presentation

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Cloud AI Privacy Vulnerability

Sending sensitive health data to cloud LLM APIs (like Gemini, OpenAI, etc.) can cause many security risks:

- **Third-Party Ingestion:** Providers may store raw health prompts for model retraining.
- **Apple Privacy Conditions:** Real-world HealthKit metadata cannot be transmitted over public external APIs.
- **Anonymization:** PII must be stripped *on-device* before hitting any cloud endpoint.

Case Study: Apple Health Export Analysis

type	sourceName	sourceVer...	unit	m_creatio...	m_creatio...	m_creatio...	m_creatio...	m_startDate
BodyMassIndex	Zepp	202306091108	count	2023-06-20	10:23:33 AM	AM	+0530	2023-06-13
BodyMassIndex	Zepp	202411071848	count	2024-11-13	05:50:10 AM	AM	+0530	2024-11-12
BodyMassIndex	Zepp	202501201440	count	2025-01-27	10:54:29 AM	AM	+0530	2025-01-27
Height	Abhishek's iPhone	16.3.1	cm	2023-03-13	07:37:56 PM	PM	+0530	2023-03-13
BodyMass	Zepp	202306091108	kg	2023-06-20	10:23:33 AM	AM	+0530	2023-06-13
BodyMass	Abhishek's iPhone	16.3.1	kg	2023-03-13	07:37:56 PM	PM	+0530	2023-03-13
BodyMass	Zepp	202411071848	kg	2024-11-13	05:50:10 AM	AM	+0530	2024-11-12
BodyMass	Zepp	202501201440	kg	2025-01-27	10:54:29 AM	AM	+0530	2025-01-27
HeartRate	Zepp	202405241932	count/min	2024-06-11	12:50:20 PM	PM	+0530	2024-06-06
HeartRate	Zepp	202405241932	count/min	2024-06-11	12:50:20 PM	PM	+0530	2024-06-06
HeartRate	Zepp	202405241932	count/min	2024-06-11	12:50:20 PM	PM	+0530	2024-06-06
HeartRate	Zepp	202405241932	count/min	2024-06-11	12:50:20 PM	PM	+0530	2024-06-06
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HeartRate	Zepp	202405241932	count/min	2024-06-11	12:50:20 PM	PM	+0530	2024-06-06
HeartRate	Zepp	202405241932	count/min	2024-06-11	12:50:20 PM	PM	+0530	2024-06-06

Analyzing the **Pacific28 Apple Health Dataset** from Kaggle reveals the complexity of "Anonymized" files.

- **Path:** /kaggle/input/datasets/pacific28/apple-health-data/health_data.csv
- **Volume:** 800,000+ records (Heart Rate, Sleep, Vitals).
- **Structure:** Deeply nested metadata including sourceName, device, and sourceVersion.

Research confirms that even without GPS data, "Hidden Metadata" can still re-identify a specific user

Identifying Hidden Leaks

The Kaggle dataset demonstrates why high-risk data extends far beyond location coordinates.

Metadata Field	Example Value	Risk Factor	MindGuard Action
sourceName	"Chelsea's Apple Watch"	**Direct Identity Link**	Mask (Presidio)
device	Serial No: 1A2B-3C4D...	**Hardware Tracking**	Redact Entirely
Timestamps	2026-08-03 14:15:22	**Routine Identification**	Generalize Time
TimeZone	GMT-0400 (EDT)	**Approx. Location**	Remove Metadata

Technical Pivot: Microsoft Presidio

Moving from cloud Gemini to a local framework makes sure there's zero network leakage/data sharing between companies

- **100% Local Processing:** Microsoft Presidio runs on the host machine. No data is sent to external servers.
- **spaCy Integration:** Uses an NLP "brain" to understand context (names, places) within mental health chat logs.
- **Custom Recognizers:** Allows writing Python rules for specific health metadata formats.

System Design: Dual-Stream Processing

Stream A: Interactive Chat

Users type directly into a Streamlit text box. Presidio redacts names and PII in real-time before displaying the output.

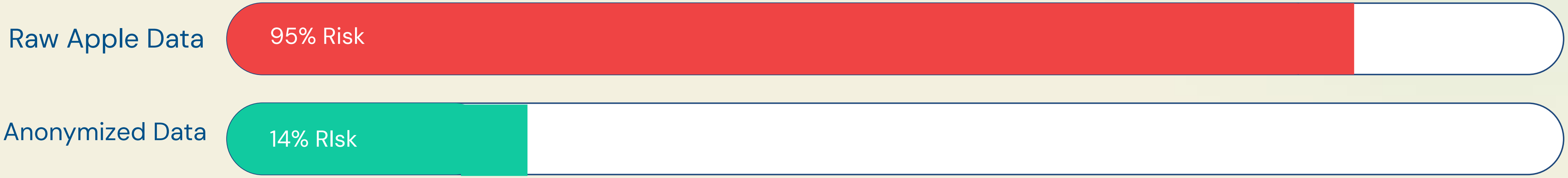
Stream B: Batch Health CSV

Users upload Apple Health export files. MindGuard parses thousands of rows (Kaggle dataset) and scrubs device identifiers.

The Privacy Risk Index

The success of the MVP will be determined by a **Privacy Risk Index (PRI)**, calculating the reduction in data sensitivity after anonymization

$$PRI = \frac{\sum \left(\text{Leakage}_{\text{Likelihood}} \times \text{Sensitivity}_{\text{Score}} \right)}{\text{Anonymization Weight}}$$



Questions

1.) Do you suggest I look at another open-source library, or do you think I should stick to my original plan of using Gemini API for now?

2.) From a privacy standpoint, do you you think I should approach the censoring with more than blurring or changing its name (like adding noise)?